

Department of Mechanical and Production Engineering

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Control Systems Notes

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1 Definitions and Notation

Let $\mathbf{x}$ denote the state vector and $\mathbf{A}$ the state matrix. A standard introduction is given by Ogata (2010).

Definition 1.1: State

The state is the smallest collection of variables that determines the future behaviour of a system when its future input is known.

Example 1.1: Second-order system

A linear state-space model may be written as

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u.$$

Theorem 1.1: State-transition matrix

For a continuous matrix $\mathbf{A}(t)$, a state-transition matrix exists.

Proof.

This follows from existence and uniqueness for linear ordinary differential equations.

As stated in Definition 1.1, the selected variables must determine the future response. See also theorem 1.1.

References

Ogata, Katsuhiko (2010). *Modern Control Engineering*. 5th ed. Prentice Hall.